Algorithm Analysis & Design

PRACTICE PROBLEMS: RECURRENCES

Name: _____ Date: _____

Part 1: The Recursion Tree Method

Determine the asymptotic bound for each of the following recurrences using the recursion tree method.

1. $T(n) = T(n/2) + n$

2. $T(n) = T(n/2) + c$ (where c is a constant)

3. $T(n) = 2T(n/2) + n$

4. $T(n) = 4T(n/2) + n$

5. $T(n) = 2T(n - 1) + 1$

Part 2: The Master Theorem Method

Solve the following recurrences using the Master Theorem. State which case applies.

1. $T(n) = 2T(n/4) + 1$

2. $T(n) = 2T(n/4) + \sqrt{n}$

3. $T(n) = 2T(n/4) + n$

4. $T(n) = 2T(n/4) + n^2$

Part 3: Changing Variables Method

Use the variable substitution method to transform and solve the recurrence below.

1. $T(n) = T(\sqrt{n}) + 1$